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Coordination chemistry: the scientific legacy of Alfred Werner†

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Alfred Werner defined the basics of coordination chemistry, work for which he was awarded the Nobel Prize in chemistry in 1913. This short article looks at his contributions and highlights how his legacy dominates coordination chemistry to this day.

1913 – the first Nobel Prize in inorganic chemistry

One hundred years ago, in 1913, Alfred Werner from the University of Zürich received the Nobel Prize in Chemistry "in recognition of his work on the linkage of atoms in molecules by which he has thrown new light on earlier investigations and opened up new fields of research especially in inorganic chemistry".[1] This was the first Nobel Prize in inorganic chemistry and the first to a Swiss chemist – so who was Alfred Werner and why are we commemorating his achievements in this volume?

Alfred Werner – the man

Personal life and background[1,2]

Alfred Werner was born on 12th December 1866 in Mulhouse in the Alsace region of France. His interest in chemistry developed while he was in his teens and during his military service Werner attended chemistry classes at the Technical High School in Karlsruhe. By the age of 20 he was studying chemistry at the

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Technical High School in Zürich (later to become the ETH Zürich) from where, in 1889, he was awarded a Diploma in Technical Chemistry. This was followed by doctoral studies in organic chemistry under the mentorship of Professor Arthur Hantzsch. The work in Werner's doctoral thesis at the University of Zürich (1890) on the spatial arrangements of atoms in nitrogen-containing compounds was to have a major influence on his ground-breaking theories of coordination compounds. Only one year later, in 1891, he submitted his Habilitation thesis entitled "*Contribution to the theory of affinity and valency*" which was evaluated as being of excellent quality and he was made a *Privatdozent* of the Technical High School. His star was truly in the ascendant after this rapid progression through the academic ranks.

After indulging in studies of thermochemistry in Paris with Marcellin Bertholet, Werner returned to Zürich, initially as a *Privatdozent* at the Technical High School. He moved to the University of Zürich in 1893 to take up a position of Associate Professor in Chemistry, primarily on the basis of his seminal publication in inorganic chemistry "*A contribution to the constitution of inorganic compounds*".[3]

Although Werner taught organic chemistry in the university, his research interests were turning increasingly towards inorganic compounds, specifically those formed between metals and ammonia (see later). In 1895, Werner was promoted to the position of full Professor of Chemistry at the University of Zürich. In the same year, he adopted Swiss nationality and married.

As a teacher, Werner was reckoned to be truly excellent, inspiring his students to study and research in chemistry through brilliant lectures and an easy personal style. In this respect, he was so successful that he soon required a larger lecture theatre to accommodate all of the students who wished to hear his lectures!

He received many honours and offers of Chairs in other Universities, including an unsuccessful attempt to attract him to our own University in 1902, but he remained in Zürich for the remainder of his career. He was a key player in the founding of the Swiss Chemical Society and was its first president in 1901. The highlight of scientific success was undoubtedly the award of the Nobel Prize for Chemistry in 1913. In 1914 he received an honorary doctorate from his *alma mater* the ETH Zürich. Alfred Werner died a few weeks before his 53rd birthday on 15th November 1919 (Fig. 1).

Fig. 1 Alfred Werner (1866–1919): the first Nobel Prize winner for work in inorganic chemistry.

Scientific contribution

Today we think of Werner as the inorganic chemist *extraordinaire*, and it is sobering to think that his training and initial studies were in the classical tradition of organic chemistry. Indeed, he did not take responsibility for the inorganic chemistry lecture courses at the University of Zürich until the early years of the twentieth century. His earliest works and publications relate to his doctoral work with his supervisor Arthur Hantsch and concern the stereochemistry of nitrogen compounds.[4] In these works, which were the key publications in extending the concept of the spatial arrangement of atoms to elements other than carbon, he demonstrated the ability to think in three dimensions that so characterised his later studies in inorganic chemistry.

His grounding in organic stereochemistry soon led him to studies of coordination compounds in which ammonia or organic amines were bound to metal centres. In this section, we will not examine all of the detailed and ultimately convincing steps that led Werner to the Nobel Prize, but rather attempt to summarise what his lasting contributions to our understanding or coordination chemistry. The fundamental step was Werner's identification of two types of "valence". The first he called a primary or ionizable valence (*Hauptvalenz*) and the other the secondary or non-ionizable valence (*Nebenvalenz*). If we substitute the modern terminology of the oxidation state and the coordination number for the terms *Hauptvalenz* and *Nebenvalenz*, we arrive at modern coordination chemistry.

He then postulated that every element tended to satisfy both its *Hauptvalenz* and *Nebenvalenz*. From this postulate, we arrive at the concept of a fixed coordination number for a particular element in a particular oxidation state.

Using this systematic approach, we can take the Werner compound that we now formulate as $[Co(NH_3)_6]Cl_3$ and assign a *Nebenvalenz* (coordination number) of six and an *Hauptvalenz* (oxidation state) of three.

Werner's toolkit – the periodic table in 1890

The fundamental toolkit for an inorganic chemist is the periodic table, and the 65 elements known in 1890 (Fig. 2) provided Werner with all that he needed to investigate the coordination chemistry of the transition metals. The major elements missing from the 1890 periodic table are the lanthanides, whose chemistry can be kindly described as being "confused" at this period, and the transuranium elements.

With the elements at his disposal, he could identify compounds with various *Nebenvalenz* and *Hauptvalenz* values, most notably platinum(II) compounds with *Nebenvalenz* four and *Hauptvalenz* two and cobalt(III) and platinum(IV) with *Nebenvalenz* six and *Hauptvalenz* three and four respectively.

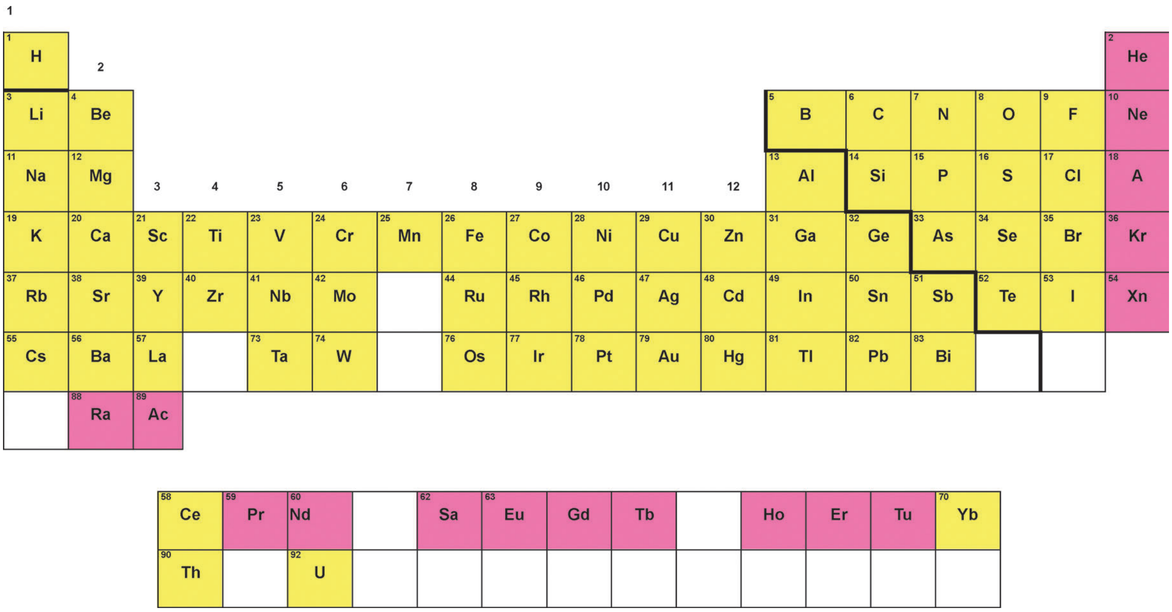


Fig. 2 The periodic table from 1890 at the beginning of Werner's scientific work (elements in yellow) and the elements in the periodic table that he published in 1905 (in pink).[5] Note the non-standardised forms of the symbols for argon (A), xenon (Xn), samarium (Sa), thulium (Tu) that he used.

Progress in the unambiguous identification of elements as well as the isolation of new ones allowed him to include 16 additional elements in his published periodic table of 1905.[5]

Werner's tools

To a modern audience, perhaps the most astonishing aspect of Werner's achievements is the minimal collection of analytical tools that he had at this disposal. The modern "standard" techniques of UV-VIS, IR, NMR and EPR spectroscopies or X-ray crystallography either did not exist, or were in their infancy and not routinely applied. Beyond the traditional tools of the chemist at that time (colour, smell, taste, melting point, solubility *etc.*), Werner also made good use of physicochemical techniques such as polarimetry and conductivity. However, ultimately Werner and his students achieved their success through impeccable and careful experimental techniques. Repeated crystallisations, subtle differences in colour or solubility, metathesis and "test-tube scale" reactions characterise the experimental work of Werner.

An aside on "valence"

The modern IUPAC definition of valence is clear and is similar in some respects to Werner's *Hauptvalenz* "The maximum number of univalent atoms (originally hydrogen or chlorine atoms) that may combine with an atom of the element under consideration, or with a fragment, or for which an atom of this element can be substituted".[6] Strictly speaking, the IUPAC definition is the *maximum* valence of the element in question and we would today state that the valence of cobalt is four (in CoF_4 (ref. 7)) but that oxidation states from −3 to +4 are routinely encountered.

However, this clarity was certainly evident in the 19th century. Frankland formulated his theory of valence in 1852 (ref. 8) in which he pulled together many strands of information from earlier work, particularly that relating to free radicals and chemical affinity, and formulated "the combining power of the attracting element... is always satisfied by the same number of atoms". Frankland developed his theory on the basis of "simple" molecules such as NCl_3, CO_2 and H_2O in which the number of combined elements corresponds to the coordination number.

These concepts were critical to the development of the structural theories of organic chemistry by Scott Couper, Crum Brown, Kekulé, le Bel and von't Hof.[9] Once again, the difference between the coordination number and the oxidation state was not so obvious with simple inorganic and organic compounds, although Kekulé found it necessary to develop a somewhat specious argument to rationalise the existence of compounds such as PCl_3 and PCl_5. His fundamental problem was an insistence that a given element could only have *one* valence (*atomicity* in the usage of Kekulé, oxidation state in modern terminology). He then defined the atomicity of phosphorus as three and called PCl_3 an *atomic compound*. Clearly, PCl_5 has an apparent atomicity of five, which is not consistent with his view that phosphorus had a valence of three *and only three*. He, therefore, introduced his concept of molecular compounds and denoted PCl_5 as $PCl_3{\cdot}Cl_2$ which served to maintain a valence of three for phosphorus and indicate the formation of PCl_5 from PCl_3 by reacting with Cl_2. Similar arguments were used to distinguish ammonia, NH_3, an atomic compound with a valence of three at the nitrogen from ammonium chloride, NH_4Cl, which was a molecular compound written $NH_3{\cdot}HCl$. This latter system still persists in the trivial nomenclature of compounds such as $[MeNH_3]Cl$ as methylamine hydrochloride. When we come to valency in inorganic chemistry, Kekulé "knew" that the valence of copper was +2. As a consequence, he formulated copper(II) sulfate pentahydrate as a molecular compound $CuSO_4\cdot5H_2O$. The dot notation of Kekulé proved convenient in coordination chemistry and in the 19th and early 20th centuries (and indeed still occasionally today) structures such as $CoCl_3\cdot6NH_3$ served both to emphasise the valence (three) of the metal centre and the overall constitution of the material *but delivered no information concerning the structure*. This is the scientific environment into which the coordination theory was launched. Werner's clear contribution was to distinguish between the contributions of charged (Cl^-, NO_2^-, SO_4^{2-} *etc.*) and neutral species (NH_3, H_2O, pyridine *etc.*).

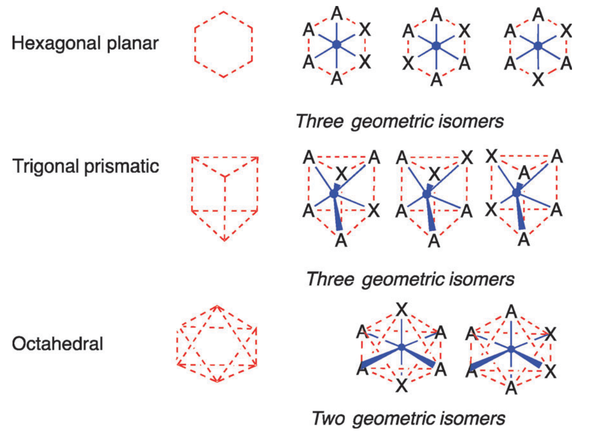
Consequences of *Nebenvalenz*

Implicit in Werner's concept of *Nebenvalenz* was not only a fixed (or variable) coordination number for a given metal centre, but also a defined spatial arrangement of the ligands. This is Werner's third postulate that the *Nebenvalenz* is directed towards fixed positions in space, in the same way as the four valences of carbon are tetrahedrally orientated. As a side note, we note that the word 'ligand' was first used by Stock in 1917 relating to silicon chemistry to extend the concept of valency from univalent partners (Cl, H *etc.*) to multiatomic radicals (CH_3, C_6H_5 *etc.*)[10] but did not gain widespread adoption in coordination chemistry until the late 1940s, in particular with its usage in the classic paper on the Irving–Williams series in 1948.[11] To return to Werner, he extended the isomer counting methods introduced by Körner to confirm the hexagonal planar structure of benzene and its derivatives to establish the octahedral spatial arrangement of the six ligands in six-coordinate cobalt(III) complexes.[12]

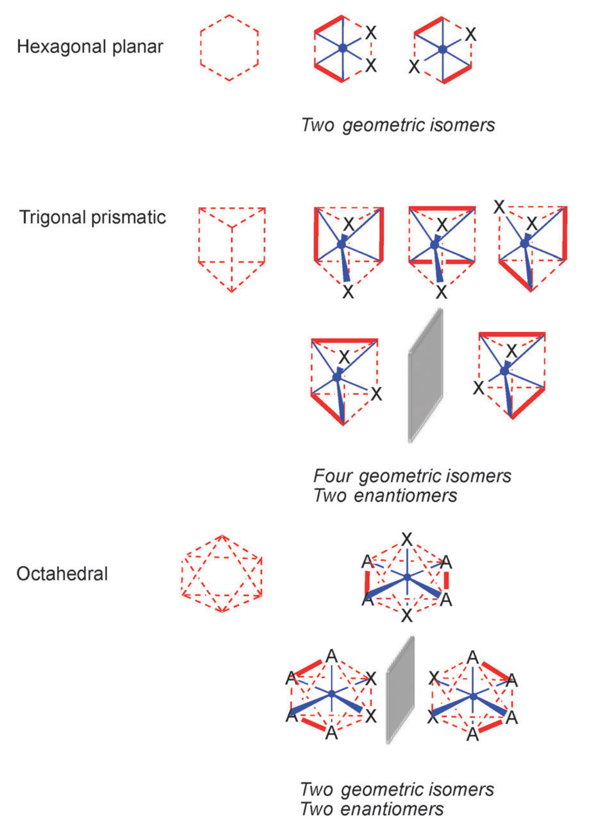
It is instructive to look briefly at how this worked in practice. It is reasonable to propose that six-coordinate metal complexes will possess hexagonal planar, trigonal prismatic or octahedral geometries. Werner considered the number of isomers that might be expected for complexes such as $[M(NH_3)_4Cl_2]$ if they adopted various geometries. For $[M(NH_3)_4Cl_2]$, one would expect three isomers for hexagonal planar or trigonal prismatic geometries, whereas an octahedral complex would give only the two *cis* and *trans* isomers (Scheme 1). In the course of extensive synthetic work in the 1890s, Werner only ever managed to isolate two isomers of $[M(NH_3)_4Cl_2]$ complexes and this provided one of the initial key pieces of support for his coordination theory and the octahedral geometry of six-coordinate complexes. This was not a conclusive proof of the geometry as it relied on the non-observation of a "missing" isomer. It was possible that Werner and his students had simply not identified or not managed to prepare the missing isomer.

Unambiguous proof came through the use of chelating ligands such as 1,2-diaminoethane (en) which played a crucial role in Werner's research and we use a complex of the type $\{M(en)_2X_2\}$ to exemplify the isomer counting analysis (Scheme 2).

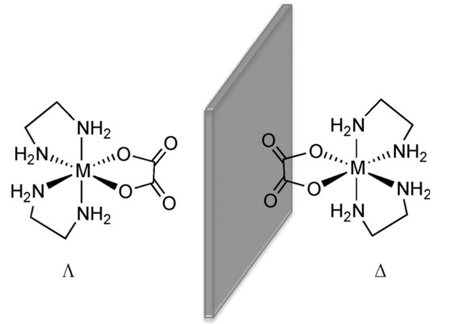
If $\{M(en)_2X_2\}$ were to possess a hexagonal planar structure, two geometrical isomers (1,2- and 1,4-arrangements of the X ligands) would be predicted (making the reasonable assumption that the bridging ligand can only link adjacent coordination sites). In contrast, if the geometry were trigonal prismatic, a total of four geometric isomers would be expected, one of which exists as a pair of enantiomers (mirror images). Finally, in the case of an octahedral structure, two geometric isomers with a *cis*- and *trans*-arrangement of the X ligands were expected, with the *cis*-compound expected to exist as a pair of



Scheme 1 The number of isomers expected for various possible geometries of a six-coordinate complex $\{M(NH_3)_4X_2\}$. The coordination polyhedron is denoted in red and the metal–ligand bonds in blue.



Scheme 2 The number of isomers expected for various possible geometries of a six-coordinate complex $\{M(en)_2X_2\}$. The coordination polyhedron is denoted in red, the metal–ligand bonds in blue and the chelating ligand in bold red. Enantiomers are presented with a grey mirror to emphasise their relationship. In the course of his work, Werner showed that only two geometric isomers could be isolated, one of which could be resolved into two enantiomers thereby establishing the octahedral geometry of the complexes.



Scheme 3 Tris chelate complexes are chiral and exist as two enantiomers which Werner managed to separate as diastereoisomeric salts with chiral anions.

enantiomers. It was the eventual isolation of both enantiomers of $\{M(en)_2X_2\}$ complexes that firmly established Werner's coordination theory.

Optical activity and chirality

A separate article in this volume discusses the enduring legacy of Werner in the field of inorganic stereochemistry,[13] but it is appropriate to consider the role of chiral compounds in his critical experiments at this juncture. Although we think of Werner as proving his coordination theory through the design and resolution of chiral compounds, it was only in 1899 (ref. 14) that he explicitly published his observations that a consequence of an octahedral coordination geometry was that tris(chelate) complexes such as $[Co(en)_2(ox)]^+$ (ox = oxalate dianion) must be chiral (Scheme 3). The experimental proof of this eluded him for another 11 years.

Werner used classical methods to resolve his cationic complexes, relying on the difference in solubility of thediastereoisomeric salts obtained with chiral anions. Denoting the enantiomers of the complex cations Δ and Λ, with a chiral anion, for example R–A, two salts can be obtained, [Δ][R–A] and [Λ][*R*–A], which in optimal cases will have different physical properties and appearance. These differences arise from the different molecular packing of the diastereoisomers in the crystal lattice, as discussed elsewhere.[41]

Bibliography and scientific output

The scientific output of Alfred Werner in his career is impressive.[15] A total of 174 research papers, 16 review articles and 46 published lectures appeared in the period 1890–1921 as well as his two exceptionally well received books *Lehrbuch der Stereochemie* (*Textbook of Stereochemistry*)[16] and *Neuere Anschauungen auf dem Gebiete der anorganischen Chemie* (*New Ideas in Inorganic Chemistry*).[17] Even today, one can learn a lot from reading these papers!

The alumni

Although Werner had an enormous influence on the development of coordination chemistry, it is probably not correct to talk about a Werner School of inorganic chemistry in Zürich or elsewhere. Coordination chemistry and organic chemistry were intimately linked at this period and the contributions of Werner had to wait for the advances in theory and characterisation techniques before they could lead to the renaissance of inorganic chemistry in the middle of the twentieth century. Nevertheless, it is quite surprising how a few of Werner's numerous doctoral students went onto academic careers. In part, the reason for this lies in the dominance of overseas students in his research group. A majority of them returned to their homelands after completing their studies in Zürich, but did not establish local schools of coordination chemistry. It is also worth mentioning at this point that Werner's research group was not only very international, but was also marked by the high number of women students, a very unusual situation at the dawn of the 20th century.

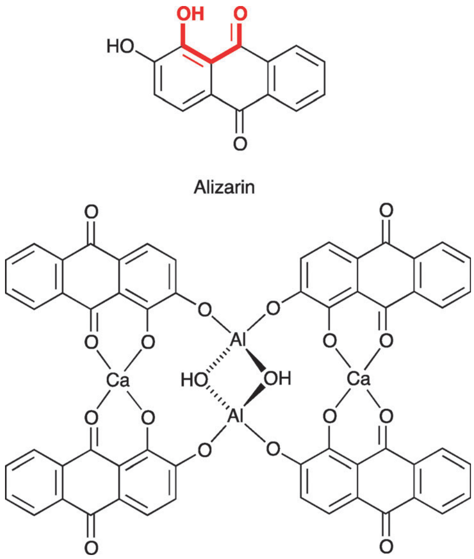
Nevertheless, a number of the alumni from Werner's laboratory did go on to lead successful academic careers, the most notable of which were probably Paul Pfeiffer and Paul Karrer. The best known is Paul Karrer who was Werner's successor as the head of the Chemistry Institute in Zürich and was a great organic chemist, receiving the Nobel Prize in chemistry himself in 1937. Probably the most successful inorganic chemist to come out of the Werner laboratory was Paul Pfeiffer, who subsequently held appointments at Rostock, then Karlsruhe, and finally Bonn. One can even see Pfeiffer as the (grand)father of supramolecular chemistry when one considers his 1915 publication "*Die Kristalle als Molekülverbindungen*" (*The crystal as a molecular compound*).[18]

Coordination chemistry before Alfred Werner

Although we rightfully regard Alfred Werner as the father of coordination chemistry, this does not mean that coordination compounds were not known before his work. In contrast, metal complexes have a long and illustrious history and it is relevant to consider the state of coordination chemistry at the time Werner commenced his scientific investigations.

Probably the earliest examples of applied coordination chemistry (although such a description would certainly not have been recognised at the time) involve dyestuffs used in antiquity. When madder root is extracted with water, a solution containing a yellow-brown dye is obtained. Attempts to dye cloth with this gives pale colours that are neither light- nor water-fast (*i.e.* they fade and the colour washes out). If the dye is combined with a *mordant* that makes it "bite" to the fabric, bright and stable colours are obtained. Typical mordants are metal salts and function by making less soluble coordination compounds which act as pigments (insoluble) or bind better to the fabric. The active dye from madder is alizarin (Scheme 4) which possesses a structure that any modern coordination chemist will recognise as containing the chelating donor set (shown in red) familiar from ligands such as pentane-2,4-dione (Hacac) and the resulting bright red pigments are believed to have multinuclear structures of the type shown in Scheme 4.[19]

Although many coordination compounds were undoubtedly obtained by the alchemists and iatrochemists, few records of specific materials from this period are known. However, some early examples of observations survive, which today we can easily interpret in terms of the formation of coordination compounds.



Scheme 4 The structure of alizarin, the dye extracted from madder and the proposed structure of the red pigment obtained after treatment with calcium and aluminium salts.

Notable amongst these is the description by Andreas Libavius of the blue colour of what we recognise as $[Cu(NH_3)_4]^{2+}$ from the reaction of the copper alloy, bronze, with aqueous ammonia (generated *in situ* from aqueous calcium hydroxide and ammonium chloride solutions):[20]

$$Cu(s) + 4NH_3 + \tfrac{1}{2}O_2 + H_2O = [Cu(NH_3)_4]^{2+} + 2OH^-$$

and the brown colour of $[Co(NH_3)_6]^{3+}$ observed by "Citizen Tassaert" when cobalt(II) salts were treated with ammonia in air.[21]

$$[Co(H_2O)_6]^{2+} + 6NH_3 + \tfrac{1}{4}O_2 + \tfrac{1}{2}H_2O = [Co(NH_3)_6]^{3+} + OH^-$$

Although these isolated observations provide a historical basis for coordination chemistry, they had no great contemporary impact and it is unlikely that the results were widely disseminated. In contrast, the coordination compound known variously as Prussian Blue, Berliner Blau and bleu prussien was well characterised, widely known and commercially important as a colour fast pigment. The compound was first prepared in 1704 by Heinrich Diesbach,[22] but detailed preparations were not published for another 20 years, indicating the commercial importance of this material to artists. Although we now recognise the material as an iron(III) hexacyanoferrate(II) of variable stoichiometry, readily prepared by the reaction of iron(III) salts with $[K_4Fe(CN)_6]$, the originally reported preparation involved the reaction of iron compounds with K_2CO_3 and nitrogen containing organic materials.[23,24]

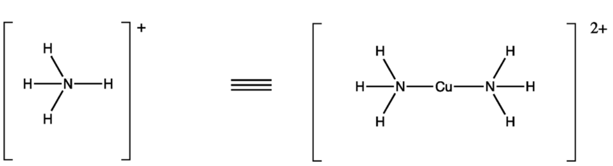
Systematic investigations in the latter part of the 19th century provided a robust experimental basis for the study of coordination chemistry and some of these will be considered in the next section where the controversy over the coordination theory will be presented. In general, the early studies were purely empirical in nature and often culminated in a series of compounds characterised by a colour and a knowledge of the empirical formula, which was often described by the name of the scientists (Zeise's salt,[25] Magnus Green salt,[26] Vaquelin's salt,[27] Peyrone's salt,[28] Reiset's second chloride,[29] *etc.*). However, to summarise the situation in 1890 when Werner began his studies, the solid body of experimental observations lacked a cohesive and inclusive model to correlate them and to deliver further understanding of the nature of these "complex"[30] materials. It is this, the all-embracing structural theory of coordination chemistry, that is the real and lasting legacy of Alfred Werner.

Controversies and the acceptance of the coordination model

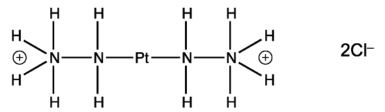
In this section we consider the alternative theories of the structures of coordination compounds which had developed in the course of the latter part of the 19th century, with a particular emphasis on the controversy between Alfred Werner and Sophus Mads Jørgensen. I am indebted to the pioneering work of George Kauffman in establishing the background to the ultimate triumph of the Werner coordination theory.[31,32]

The discussion in this section will concentrate on the ammine complexes which were both the best known and best characterized coordination compounds and at the core of Werner's research.

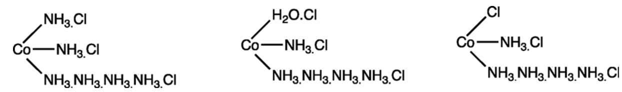
One of the earliest theories that was proposed for the structure of the metal ammine complexes was from Thomas Graham better known for his achievements in physical chemistry.[33] Graham proposed that the ammine complexes could be regarded as derivatives of the ammonium ion in which hydrogen atoms had been replaced by metal centres. Graham exemplified this theory with the structure of the diamminecopper(II) in which a hydrogen atom from each of the two ammonium ions is replaced by a single copper (Scheme 5). This satisfies the valence of copper (2) and accounts for the non-basic character of coordinated ammonia. The theory has a number of problems associated with it, the most substantial one is that the number of ammonia molecules associated with a metal can only equal the valence. The problem of Graham's copper complex is compounded by Dalton's incorrect assumption that the formula of water was OH; this in turn led to errors in atomic and molecular weights and incorrect stoichiometries for compounds. Grahams "diamminecopper(II)" was actually the well-known $[Cu(NH_3)_4]^{2+}$ cation and with this stoichiometry, the valence of copper is not satisfied by the formation of *four* Cu–N bonds. Like all good theories, the basic postulate was modified over time by others to accommodate a variety of experimental observations and eventually abandoned



Scheme 5 The Graham ammonium model for coordination complexes replaced hydrogen atoms of the ammonium ion with metal centres.



Scheme 6 The Berzelius conjugate theory proposed a structure for $[Pt(NH_3)_4]Cl_2$ in which nitrogen atoms formed chains with N–N bonds, analogous to C–C bonds in organic compounds.



Scheme 7 Jørgensen representations of $[Co(NH_3)_6]Cl_3$ (three ionic chlorides), $[Co(NH_3)_5(H_2O)]Cl_3$ (three ionic chlorides) and $[Co(NH_3)_5Cl]Cl_2$ (two ionic chlorides). Note that the need to catenate the nitrogen atoms of the ammonia is predicated upon the coordination number and the oxidation state of the metal being the same, in this case, three.

when the Werner model provided a better global explanation for the data. A major factor in the abandonment of the modified Graham model was Jørgensen's observation that compounds analogous to the ammine complexes could be prepared with tertiary amines such as Me_3N, which contained no replaceable hydrogen atoms. Even this argument is not so convincing with the benefit of hindsight, since Me_3NH^+ *does* contain such a hydrogen.

The conjugate theory subsequently proposed by Berzelius has little merit beyond the linguistic ambiguity of the copulation of metal centres and ammonia ligands and had been largely abandoned by the time Werner commenced his work. The basis was to extend the concepts of organic chemistry in which structures were formed by the C–C bonds to those complexes with multiple ammine ligands, invoking the formation of N–N bonds. Once again, the fundamental problems related to the valence of the nitrogen atoms (Scheme 6).

An interesting precursor to the Werner theory is that from Carl Claus (also known as Karl Klaus), who addressed the observations that the coordinated ammonia ligands were non-basic and formulated three postulates (recast in modern terms):[34]

(1) If ammonia combines with metal chlorides, neutral substances are formed, in which the ammonia is not basic and the ammonia cannot be detected by the usual methods.

(2) If the chlorine is replaced by hydroxide, strong bases are obtained, whose basicity is independent of the number of ammonia ligands.

(3) The number of ammonia ligands is not random one; the number of water ligands in hydrates is similar to the number of ammonia ligands in the ammines.

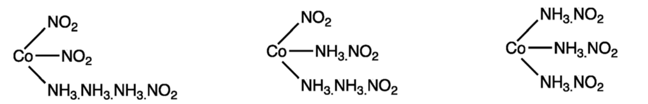
All of these postulates were attacked at various points and this model, which actually contains many of the features of Werner's coordination theory, was replaced by the Blomstrand–Jørgensen theory. The latter model is conceptually related to that of Berzelius as it is predicated upon the catenation of nitrogen atoms to form N–N bonds analogous to C–C bonds. The key concept was the concatenation of *ammonia* molecules to give $-NH_3-NH_3-NH_3-$ units, in which each nitrogen was explicitly five coordinate. The modern distinction between the coordination number and the oxidation state comes to our aid here: nitrogen can have an oxidation state of five, for example in nitrate ions, but *not* a coordination number of five. The need to catenate the ammonia ligands was based on the concept of a fixed valence for the metal – once again, the assumption that the oxidation state and the coordination number had to be the same. The Jørgensen representations of some cobalt(III) ammines are shown in Scheme 7.

It is important to stress that the painstaking experimental work of Jørgensen provided the basis for both his own and for Werner's theories. Although the theory of Jørgensen has been consigned to history, the pioneering experimental studies remain critical to the development of modern coordination chemistry.

This model accounted for various observations, although it did not provide an adequate theoretical basis for the postulates. For example, it was known that in various complexes some but not all of the chlorine present could be precipitated as AgCl with silver nitrate. Today, we explain this in terms of coordinated chloride ligands and ionic chloride counterions. The Blomstrand–Jørgensen model also equated the chlorine that could not be precipitated with chloride ligands directly attached to a metal and the ''ionic'' chloride as being associated with the end of a chain of ammonia molecules. This is seen in Scheme 6, where $[Co(NH_3)_6]Cl_3$ and $[Co(NH_3)_5(H_2O)]Cl_3$, with three N–Cl or two N–Cl and one O–Cl ''bonds'', respectively, have three chlorides which can be precipitated as AgCl, whereas $[Co(NH_3)_5Cl]Cl_2$ has one chloride ligand directly bonded to the metal and so only two ionic chlorides can be precipitated as AgCl.

We now consider the cobalt(III) complexes of the type $[Co(NH_3)_6]A_3$, $[Co(NH_3)_5A]A_2$, $[Co(NH_3)_4A_2]A$ and $[Co(NH_3)_3A_3]$ where A is an anionic species (most often halide or nitrite in the classical studies). The Werner and Jørgensen models coincide in their predictions for the $[Co(NH_3)_6]A_3$, $[Co(NH_3)_5A]A_2$ and $[Co(NH_3)_4A_2]A$ complexes. In addition to the data obtained from reactivity studies with silver nitrate, Werner also studied the conductivity of solutions of these complexes, which gave rise to four, three and two ions in solution respectively. The critical difference between the two models arises with the complexes $[Co(NH_3)_3(NO_2)_3]$. According to the possible Jørgensen structures, the complex will always give rise to conducting solutions with two, three or four ions in solution. In contrast, the Werner model predicts a non-conducting molecular species in which all of the nitrito ligands are coordinated to the metal. Solutions of the complex $[Co(NH_3)_3(NO_2)_3]$ were non-conducting, a critical observation which eventually led to the dominance of the Werner coordination theory over the Jørgensen model. (Scheme 8).[35,36]

It is useful to summarise the difference between the two models. The Jørgensen approach was based on the assumption that the coordination number of the cobalt could not be greater than three, as defined by the oxidation state of the metal. In contrast, Werner allowed the coordination number to vary from the oxidation state.



Scheme 8 The possible Jørgensen structures for $[Co(NH_3)_3(NO_2)_3]$. In all of the formulations there is at least one ionic nitrite and the complex is predicted to be an ionic conductor in solution.

Major scientific contributions of Werner

From then to now – Werner's scientific heritage

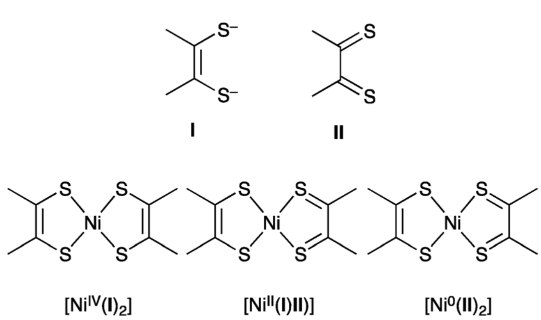
Primary and secondary valence. As we have seen, Werner's concepts of *Hauptvalenz* and *Nebenvalenz* have stood the test of time in their modern incarnations of the oxidation state and the coordination number. Naturally, in the course of the last 100 years, a number of new complexities have emerged which challenge the simplicity of the original concept.

Although the oxidation state is usually clear with classical Werner-type ligands, so-called *non-innocent* ligands lead to the oxidation state of the metal centre being ambiguous. This ambiguity arises from the ligands themselves being redox active and the term arises from an early publication of (Axel) Christian Jørgensen in which he described ligands as *innocent* or *suspect*. The simplest non-innocent ligand is NO, which in a neutral-atom electron counting formalism can act as a one-electron or a three-electron donor or in the charged formalism as NO^+ or NO^-. The formalism leads to an oxidation state ambiguity of two at the metal, which is usually resolved by electron counting (for example, using 18-electron rule) or by crystallography. The latter tool, not available to Werner, allows a ready distinction between the one electron (NO^-) formalism which has a bent M–N–O structure and the three electron (NO^+) formalism which is linear.

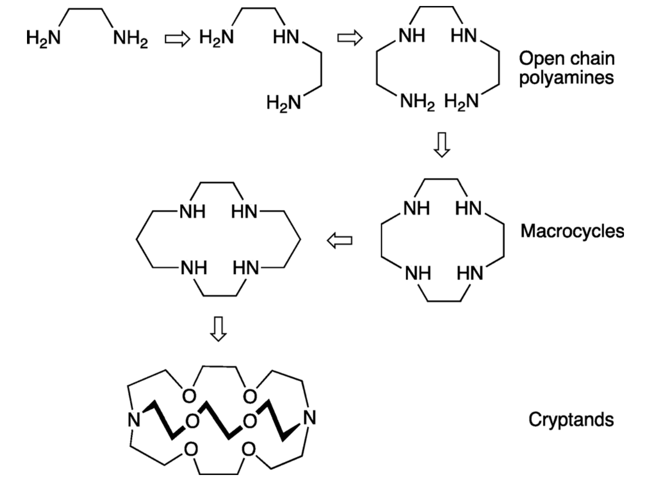
Other classical examples of non-linear ligands include the dithiolenes (Scheme 9), where the ambiguity in the oxidation state of the ligand leads to limiting formulations as neutral or dianionic species.

Metal compounds are now routinely encountered with oxidation states between −2 ($[Fe(CO)_4]^{2-}$) and +8 (OsO_4).

Ligands. The key to modern coordination chemistry is the ability to design ligands with particular properties to enforce a



Scheme 9 Dithiolenes are typical examples of non-innocent ligands. The neutral bis(thione) form **II** is related to the bis(thiolate) form **I** by a two electron reduction. The binding of two ligands to a nickel centre leads to an ambiguity in the oxidation state of the metal (from Ni(IV) to Ni(0)) depending on the oxidation state of the ligand.



Scheme 10 The ''simple'' 1,2-diaminoethane ligand has expanded and increased in complexity over the years giving new families of open chain, cyclic and encapsulating ligands.

particular coordination mode onto a metal centre. The origins of ligand design lie in Werner's recognition of the stereogenic consequences of chelating ligands such as 1,2-diaminoethane in his classical studies leading to the isolation of chiral complexes.

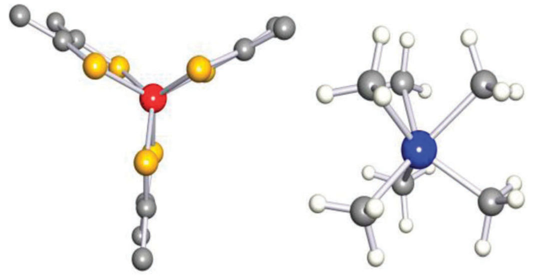
1,2-Diaminoethane is the first of a series of ligands which leads conceptually to macrocycles and cryptands (Scheme 10).

It is interesting to note that the development of the oxygen analogs of these species (the crown ethers) and also the encapsulating ligands such as the cryptands is recognised by the award of the 1987 Nobel Prize in Chemistry to Jean-Marie Lehn, Donald Cram and Charles Pedersen. In passing, we note that Jean-Marie Lehn was born in Rosheim in the Alsace, only 80 kilometres from Werner's birthplace in Mulhouse.

Coordination geometry and stereochemistry. A separate article in this volume[13] is concerned with the influence that Werner and his ideas had on inorganic stereochemistry. The extension of the idea of a three-dimensional structure for metal complexes combined with the concepts of *Haupt-* and *Nebenvalenz* truly revolutionised coordination chemistry.

Although Werner was concerned primarily with *Nebenvalenz* of six and four, the design of ligands which force metal ions to adopt unusual coordination numbers and geometries has exercised the minds and fingers of inorganic chemists since his time. The role of this article is not to provide a comprehensive overview of the advances in this area in the past 100 years, but rather to identify a few highlights which would probably have appealed to Werner.

Although Werner's classical proofs of the octahedral coordination geometry of cobalt(III) and the majority of other six-coordinate complexes has been amply confirmed by X-ray structural determination, they do not preclude the occurrence of trigonal prismatic or even planar complexes. Trigonal prismatic compounds such as the dithiolene complex $[Re(Ph_2C_2S_2)_3]$[37] are now well established (Scheme 11) and the factors leading to this geometry are well-understood. Even the ''simple'' complex $[W(CH_3)_6]$[38] exhibits a trigonal prismatic geometry.

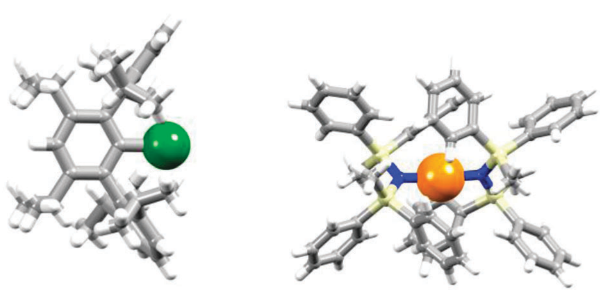


(a) (b)

Scheme 11 The trigonal prismatic structures of (a) [Re($Ph_2C_2S_2$)$_3$] (H atoms are omitted) and (b) [W(CH_3)$_6$].

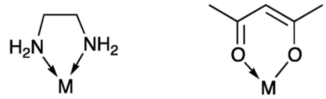
Nowadays, very low and high coordination numbers are known but would have been unexpected to Werner. For the very rare case of one-coordination, we have to look at main group organometallic chemistry to find, for example, {C_6H-2,6-(C_6H_3-2,6-$^iPr_2$)2-3,5-$^iPr_2$}Ga (Scheme 12a)[39] in which extreme steric hindrance protects the metal atom. Gold(I) is typically two-coordinate (linear) but in [Fe{N(SiMePh$_2$)$_2$}$_2$], it is the steric crowding of the ligands that restricts the number of coordinated ligands (Scheme 12b).[40] High coordination numbers for d-block metal ions are often associated with the second and third row metals combined with simple monodentate or simple chelating ligands, for example, $[Mo(CN)_8]^{3-}$, $[W(CN)_8]^{4-}$, $[ReH_9]^{2-}$, [Mn(NO$_3$-*O*,*O*′)$_4$]$^{2-}$ (8-coordinate) and $[Cr(O_2)_4]^{3-}$ (8-coordinate). In [M(BH$_4$)$_4$] and (M = Hf, Zr), each $[BH_4]^-$ is tridentate leading to 12-coordination.

The chelate effect. The chelate effect and its phenomenological child, the macrocyclic effect, lie at the core of modern ligand design. Polydentate or multidentate ligands have two or more donor sites and when they can bind to a single metal centre generating new heterocyclic rings (Scheme 13) the ligands are described as chelating and the structure is described as a chelate. The modern emphasis is on the *stability* of chelate complexes and it is a general observation that they are thermodynamically more stable than complexes containing an equal number of comparable monodentate ligands. As always, the devil lies in the detail and in quantifying the stability, good and appropriate model compounds need to be defined. The impact on ligand design is clear – to get more stable complexes, chelating ligands will be better than monodentate ligands.



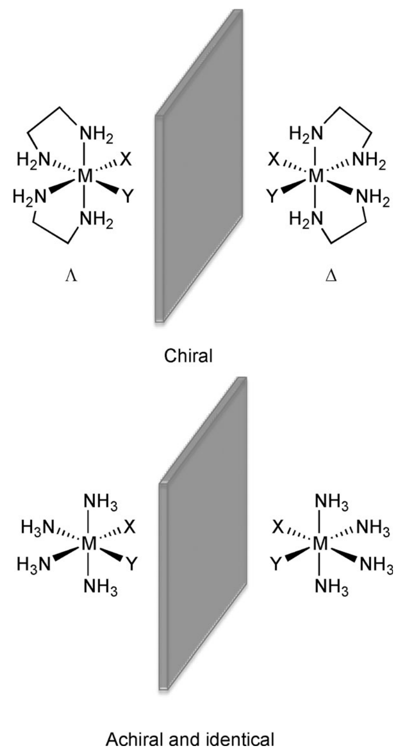
(a) (b)

Scheme 12 (a) {C_6H-2,6-(C_6H_3-2,6-$^iPr_2$)$_2$-3,5-$^iPr_2$}Ga is a very unusual example of one-coordination; (b) Fe(II) is two-coordinate in [Fe{N(SiMePh$_2$)$_2$}$_2$].



Scheme 13 Two examples of chelating ligands in which two donor atoms of a polydentate ligand are coordinated to a single metal centre. These examples, comprising 1,2-diaminoethane (en) and pentane-2,4-dionato (acac) ligands also illustrate a number of general points; the chelating ligands may be neutral or charged (most commonly anionic), the size of the heterocyclic ring may vary but is most commonly five or six-membered, the atoms constituting the heterocyclic chelate ring may be saturated or unsaturated.

Ligands such as en (introduced as a ligand by Jørgensen) and pn, as well as *O*,*O*′-donors such as [acac]$^-$ played a critical role in the studies of Alfred Werner although the first systematic studies and syntheses of chelate complexes should be attributed to Jørgensen.[41] Werner clearly recognised the stereogenic consequences of these chelate rings and, indeed, complexes such as [Co(en)$_2$X$_2$]$^+$ (X = Cl, Br *etc.*) played a critical role in the establishment of his coordination theory. Summarising much detailed work, complexes such as *cis*-[Co(en)$_2$X$_2$]$^+$ are chiral with two non-superimposable mirror image forms whereas the ''parent'' complexes [Co(NH$_3$)$_4$X$_2$]$^+$ are achiral (Scheme 14).[13] For a more detailed discussion, the reader is referred elsewhere in this volume. However, although Werner clearly recognised the stereogenic consequences arising from chelating ligands and almost certainly had a qualitative feel for the enhanced stability of their complexes, he was not actively concerned with quantification of the stability. Indeed, the basic



Scheme 14 *cis*-[Co(en)$_2$X$_2$]$^+$ is chiral with two non-superimposable mirror image but the ''parent'' complex [Co(NH$_3$)4X$_2$]$^+$ is achiral.

principles of determining stepwise stability constants were only formulated in the 1914–1916 period by Jaques[42] and Bjerrum[43] and systematic studies of the stability of chelate compounds were not routine until 1945 when Schwarzenbach[44] and Calvin (later to receive the Nobel Prize in Chemistry for his discovery of the Calvin cycle in photosynthesis)[45] commenced their pioneering work. The excitement and the pace at which understanding was developing in this renaissance of coordination chemistry is well captured in a contemporary monograph on chelate compounds (which is, incidentally, dedicated to Werner's student Paul Pfeiffer).[46]

As mentioned earlier, the stablest chelate rings are typically five- or six-membered, a fact that was recognised by Ley[47] and Chugae. The description of *chelating* for such ligands and the use of chelate to describe the resultant complexes was first introduced in 1920 by Morgan and Drew,[48] who related the binding of the two donor atoms to the metal as analogous to a Crab's claw (Greek *chela,* $\chi\eta\epsilon\lambda\alpha$) gripping an object.

Werner and the periodic table

Although we do not often think of Werner as one of the pioneers in the development of the periodic table, he was one of the first authors to clearly distinguish between the transition and main group elements and to assign the transition elements to the central part of the periodic system. In a paper from 1905,[49] he clearly distinguishes between the transition elements and the main group elements and solves the problem that had hitherto led to the grouping of elements such as chromium and sulphur (both of which form XO_3 as the highest oxides) together. The periodic system he produced very closely resembles our modern table, even to the extent of including the lanthanides (when known) in their customary position of the long form of the table. It is worth noting that Werner came to his construction of the periodic table on the basis of a sound understanding of the *chemistry* of the elements and not on the basis of quantum theory and the electronic structure of the atoms. It is remarkable that Werner left space for 15 elements in the f-series in the absence of knowledge of electronic structure.

It is interesting that the theoretical and experimental basis for the placing of the lanthanides in the periodic table as a subgroup was only predicted by Bohr in the final years of Werner's life, work for which he also received the Nobel Prize (in Physics) in 1922.[50] Even the number of lanthanide elements was hotly debated, with claims for something like 100 rare-earth elements being made in the latter half of the 19th century. Most of these were mixtures and it required the pioneering work of Henry Moseley (who tragically died during the Battle of Gallipoli in 1915) using X-ray spectroscopy to establish that there were exactly 15 (and no more) lanthanides.[51]

Chemists are people too! The history of chemistry as a teaching tool

We have said little about the personal life of Alfred Werner in this article, but suffice it to say that he was a man with all of the virtues and faults one might expect and with a tendency to overindulge! In our teaching of chemistry today we concentrate on facts and the results obtained and have little or no emphasis on the scientists who lie behind these observations.

This is understandable, but leads to a tendency to dismiss as irrelevant all that went before or is not in the last five year's literature. To quote Marcus Tullius Cicero, *Nescire autem quid antequam natus sis acciderit, id est semper esse puerum*. (Not to know what happened before you were born, is to be a child forever.)[52]

Does it matter in the global scheme of things, that Lavoisier married his wife when she was thirteen, that she was his dedicated coworker and played a major role in establishing his reputation after his execution? Possibly not. But, is it interesting and does it make one wish to learn more about Antoine and Marie-Anne Lavoisier? We think so.

Is it important to know that the Fahrenheit temperature scale, still in use in parts of the world and with the freezing point of water at +32 F and the boiling point of water at +212 F, is based on the lowest stable temperature that Fahrenheit could achieve (a 1 : 1 : 1 mixture of ice, water, and ammonium chloride) and the temperature of his wife's armpit? Probably not. Is it interesting? We think so.

Do we need to know that the cigar box that Glenn Seaborg used to contain the first tangible samples of plutonium was borrowed from his colleague Gibert N. Lewis, who postulated the covalent bond and who is commemorated in the Lewis structures that we still use? Maybe not, but it all brings the human element and the magnitude of the achievements into perspective.

The scientists behind the stories, the successes, the struggles and indeed the scandals belong to our scientific education as much as the raw facts. We hope that in this article, we have managed to show how one man has influenced and dominated our understanding of inorganic chemistry for the past one hundred years.

Notes and references

1 http://www.nobelprize.org/nobel_prizes/chemistry/laureates/1913/.
2 G. B. Kauffman, in *Coordination Chemistry. A Century of Progress*, ed. G. B. Kauffman, ACS Symposium Series, Washington, 1994, 565, p. 3.
3 A. Werner, *Z. Anorg. Allg. Chem.*, 1893, **3**, 267.
4 A. Hantzsch and A. Werner, *Ber.*, 1890, **23**, 11.
5 A. Werner, *Ber.*, 1905, **38**, 914.
6 http://goldbook.iupac.org/V06588.html.
7 J. V. Rau, S. N. Cesaro, N. S. Chilingarov and G. Balducci, *Inorg. Chem.*, 1999, **38**, 5695.
8 E. Frankland, *Philos. Trans. R. Soc. London*, 1852, **142**, 417.
9 P. J. Ramberg, *Chemical Structure, Spatial Arrangement: The Early History of Stereochemistry, 1874–1914*, Ashgate Publishing, Farnham, 2003.
10 A. Stock, *Ber.*, 1917, **50**, 170.
11 H. Irving and R. J. P. Williams, *Nature*, 1948, **162**, 746.
12 W. Körner, *Gazz. Chim. Ital.*, 1870, **4**, 305.
13 E. C. Constable, *Chem. Soc. Rev.*, 2013, DOI: 10.1039/C2CS35270B.
14 A. Werner and A. Vilmos, *Z. Anorg. Chem.*, 1899, **21**, 145.
15 G. B. Kauffman, *Alfred Werner. Founder of Coordination Chemistry*, Springer-Verlag, Berlin, 1966.
16 A. Werner, *Lehrbuch der Stereochemie*, Gustav Fischer, Jena, 1904.
17 A. Werner, *Neuere Anschauungen auf dem Gebiete der anorganischen Chemie*, Friedrich Vieweg, Braunschweig, 1905.

18 P. Pfeiffer, *Z. Anorg. Allg. Chem.*, 1915, **92**, 376.
19 C.-H. Wunderlich and G. Bergerhoff, *Chem. Ber.*, 1994, **127**, 1185.
20 A. Libavius, D.O.M.A. Commentationum Metallicarum Libri Quatuor de Natura Metallorum, Mercurio Philosophorum, Azotho, et Lapide seu tinctura physicorum conficienda è Rerum Natura, Experientia, et Autorum præstantium fide, Frankfurt, 1597.
21 Tassaert, *Ann. Chim. Phys.*, 1798, **28**, 92.
22 J. L. Frisch, *Miscellanea Berolinensia ad incrementum Scientiarum*, 1710, **1**, 377.
23 J. Brown, *Philos. Trans. R. Soc. London*, 1724, **33**, 17.
24 J. Woodward, *Philos. Trans. R. Soc. London*, 1724, **33**, 15.
25 W. C. Zeise, *Annu. Rev. Phys. Chem.*, 1831, **97**, 497.
26 G. Magnus, *Annu. Rev. Phys. Chem.*, 1828, **14**, 239.
27 L.-N. Vauquelin, *Ann. Chim.*, 1813, **88**, 188.
28 M. Peyrone, *Ann. Chem. Pharm.*, 1844, **51**, 1.
29 J. Reiset, *Compt. Rend.*, 1844, **18**, 1100.
30 A. Werner, *Z. Anorg. Chem.*, 1893, **3**, 267.
31 G. B. Kauffman, in *Comprehensive Coordination Chemistry I*, ed. G. Wilkinson, R. D. Gillard and J. A. McCleverty, Pergamon, Oxford, 1987, p. 1.
32 G. B. Kauffman, *J. Chem. Educ.*, 1974, **51**, 522.
33 T. Graham, *Elements of Chemistry*, Baillike, London, 1837.
34 C. Claus, *Ann. Chim.*, 1856, **98**, 317.
35 A. Werner and A. Miolati, *Z. Phys. Chem.*, 1893, **12**, 35.
36 A. Werner and A. Miolati, *Z. Phys. Chem.*, 1894, **14**, 506.
37 R. Eisenberg and J. A. Ibers, *Inorg. Chem.*, 1966, **5**, 411.
38 V. Pfennig and K. Seppelt, *Science*, 1996, **271**, 626.
39 Z. Zhu, R. C. Fischer, B. D. Ellis, E. Rivard, W. A. Merrill, M. M. Olmstead, P. P. Power, J. D. Guo, S. Nagase and L. Pu, *Eur. J. Chem.*, 2009, **15**, 5263.
40 R. A. Bartlett and P. P. Power, *J. Am. Chem. Soc.*, 1987, **109**, 7563.
41 S. M. Jørgensen, *J. Prakt. Chem.*, 1889, **39**, 1.
42 A. Jaques, *Complex ions in aqueous solutions*, Longmans, Green and Co, London, 1914, http://openlibrary.org/books/OL7145044M/Complex_ions_in_aqueous_solutions.
43 N. Bjerrum, *Fys. Tidskr.*, 1916, **16**, 59.
44 G. Schwarzenbach, *Helv. Chim. Acta*, 1945, **28**, 828.
45 M. Calvin and K. W. Wilson, *J. Am. Chem. Soc.*, 1945, **67**, 2003.
46 A. E. Martell and M. Calvin, *Chemistry of the metal chelate compounds*, Prentice-Hall, New York, 1952.
47 H. Ley, *Z. Elektrochem.*, 1904, **10**, 954.
48 G. T. Morgan and H. D. K. Drew, *J. Chem. Soc.*, 1920, 1456.
49 A. Werner, *Ber.*, 1905, **38**, 914.
50 N. H. D. Bohr, *Philos. Mag.*, 1913, **1**, 476–857.
51 H. G. J. Moseley, *Philos. Mag.*, 1913, 1024.
52 M. T. Cicero, *De Oratore, Book 3*, ed. D. Mankin, Cambridge University Press, Cambridge, 2011.